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SUPERSEDING
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January 9, 1934

FEDERAL SPECIFICATION

HANDLES, ASH, FOR TOOLS

This specification was approved on the above date by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers ash handles for use in post-hole diggers, forks, hoes, rakes, and shovels as described in Federal Specifications GGG–D–326, GGG–F–601, GGG–H–496, GGG–R–96, and GGG–S–326, respectively. (See 2.1.)

1.2 Types, classes, and styles. — The handles shall be of the following types, classes, and styles, as specified:

Diggers, post-hole (one type only).

Forks:
- Type I.—Hay fork:
  - Class A.—Bent long handle.
  - Class B.—Straight long handle.
- Type II.—Manure fork:
  - Class A.—Bent long handle.
  - Class B.—Bent D-handle.
- Type III.—Mill fork:
  - Class A.—Bent D-handle.
- Type IV.—Spading fork:
  - Class A.—Bent long handle.
  - Class B.—Bent D-handle.

Hoes:
- Type I.—Garden hoe.
- Type II.—Mortar hoe.
- Type III.—Mortar mixing hoe.

Rakes:
- Type I.—Asphalt (tar).
- Type II.—Garden:
  - Class A.—Bow.
  - Class B.—Level head.
- Type III.—Lawn:
  - Class A.—Steel, flexible head:
    - Style 1.—Full radial head.
    - Style 2.—Semiradial head.
  - Class B.—Wood.
- Type IV.—Road.

Shovels:
- Type I.—D-handle:
  - Class A.—17- to 20-inch lift.
  - Class B.—9½- to 12½-inch lift (garden spade).
  - Class C.—5½- to 8½-inch lift (ditch and drain spade).
- Type II.—Long handle:
  - Class A.—17- to 20-inch lift.
  - Class B.—9½- to 12½-inch lift (garden spade).
  - Class C.—5½- to 8½-inch lift (ditch and drain spade).
- Type III.—Telegraph handle:
  - Class A.—Straight handle.
    - 8-foot.
    - 10-foot.
  - Class B.—Bent handle.
    - 8-foot.
    - 10-foot.

1.3 Grades.—The handles shall be of the following grades, as specified (see table I):

Grade A.—For forks, hoes, and rakes.
Grade SA.—For shovels.
Grade B.—For forks, hoes, and rakes.
Grade SB.—For shovels and post-hole diggers.

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TABLE I

The quality requirements given herein conform to Simplified Practice Recommendation R76–40, Ash Handles, for grades A, SA, B, and SB except that weight limitations are based on a moisture content not exceeding 12 percent instead of 8 percent. Long handles and stems for forks, rakes, hoes, and kindred farm and garden tools are described by grades A and B; long handles and stems for shovels, spades, and scoops, by grades SA and SB. A stem is a short handle used to make D handles. The grading of stems conforms substantially to that specified for the lower two-thirds of corresponding long handles.

Grade	Color	Number of annual rings per inch of radius		Weight in pounds per cubic foot[1]	Maximum permissible cross grain[2]		Admissible blemishes	
		Minimum	Maximum		For stems	For long handles	Kind	Number
A	White, light red, and light brown; dark red and brown in 10 inches at both ends.[3]	5	17	Over 44 (heavy)	½ diameter in full length	½ diameter in bottom ⅔, ½ diameter in top ⅓, ⅞ diameter in top 7 inches	Slight dip grain (not more than 1 per handle), pin twig marks, small streaks, and slight butterfly marks[4]	One for each 24 inches of handle length of fraction thereof, at least 12 inches apart.
SA	White, light red, and light brown; dark red and brown in 10 inches at tool end.	5	15	do	do	do	do	Do.
B	White, red, brown, and mildly variegated (calico); dark red and dark brown in 10 inches at both ends.[3]	3	21	Over 38 to 44 (medium)	1 in 12 anywhere in the full length	1 in 12, 1 in 6 in 12 inches at one end	Small knots, if located at the ends, and slight dip grain (not more than two of each per handle), pin twig marks medium streaks, medium butterfly marks, and wavy grain[4 5]	Two for each 18 inches of handle length or fraction thereof, at least 6 inches apart.
SB	White, red, and brown; dark red and dark brown in 10 inches at tool end.	4	18	do	Full diameter in any 30 inches of length	Full diameter in any 30 inches of bottom ⅔, ¾ diameter in top ⅓, ⅝ diameter in top 7 inches	Slight dip grain, pin twig marks, medium streaks, medium butterfly marks, and wavy grain[4]	Do.

(See next page for footnotes to above table.)

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2. APPLICABLE SPECIFICATIONS

2.1 The following Federal Specifications, of the issues in effect on date of invitation for bids, form a part of this specification:

QQ-S-781 —Strapping, Flat; Steel.

GGG-D-326—Diggers; Post-Hole.

GGG-F-601—Forks; Hay, Manure, Mill, and Spading.

GGG-H-496—Hoes; Garden and Mortar.

GGG-R-96 —Rakes; Hand.

GGG-S-326—Shovels (Scoops, Spades, and Spoons).

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

2.2 Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.

3.1.1 *Species of wood.*—The handles shall be made of white ash.[1]

3.1.2 *Seasoning.*—The moisture content of the handles shall not exceed 12 percent of the oven-dry weight.

3.2 Quality.—The handles shall conform to the quality requirements of table I for the grades specified.

3.3 Size and form.—The handles shall be formed and shaped to fit properly the tools described in the appropriate Federal specifications listed in 2.1 and shall be of the diameters and lengths given therein.

3.4 Workmanship.—The handles shall be smoothly finished and shall be of good commercial workmanship.

[1] "White ash" as used in this specification includes wood cut from the following species: white ash (*Fraxinus americana*), green ash (*Fraxinus pennsylvanica var. lanceolata*), and blue ash (*Fraxinus quadrangulata*).

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[1] Based on a moisture content not exceeding 12 percent.

[2] Deviation of the fiber from a line parallel with the axis of the handle. For example: ½ diameter in bottom ⅔ means that the grain may cross ½ the diameter of the handle in not less than ⅔ the length, measured from the bottom.

[3] Dark red and dark brown permitted on stems at tool end only.

[4] Blemishes are defined and classified as follows:

Butterfly mark.—Discoloration resembling somewhat the trail left by a snail.

Medium butterfly mark.—In contrast with the color of the wood.

Slight butterfly mark.—Matching the general color of the wood.

Pin twig mark.—Mark made by a very small twig.

Streak.—Thread-like discoloration.

Medium streak.—1 to 4 inches in length.

Small streak.—Not more than 1 inch in length.

Dip grain (knurl).—Grain dipping into and out of the handle, indicated by a fairly oval contour of grain on the surface.

Slight dip grain.—Not less than 8 inches long, nor deeper than ⅛ the minimum diameter of the handle at the point where the dip occurs.

Small knot.—Average diameter not more than ⅛ inch.

Admissible blemishes shall not be such as to impair the strength or serviceability of the handle.

[5] Knots will be permitted in stems at tool end only.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—At least one handle from each 24 handles or fraction thereof delivered shall be taken for inspection and tests.

4.2 Inspection and tests. — Handles taken for inspection shall be measured and inspected visually. Ordinarily it will not be necessary to weigh, count the annual rings, or determine the moisture content of each handle inspected. However, such detailed inspection shall be made in all cases where there is doubt as to the conformance of the handles with the requirements of this specification.

4.2.1 *Dimensions.*—Suitable measuring instruments such as calipers, a graduated rule, etc., shall be used in determining the dimensions of the handles.

4.2.2 *Moisture content.*—When it is necessary to determine the moisture content of handles, either of the following methods shall be used.

4.2.2.1 *Oven-drying method.*—Cut a section 1 inch long, not less than 6 inches from either end of the handle. Weigh immediately on a scale that is accurate to ½ of 1 percent. This is the original weight. Dry in an oven maintained at a temperature of 212° to 221° F. (100° to 105° C.) until constant weight is attained. Reweigh the sample. This is the oven-dry weight. Compute moisture content by the following formula:

$$\frac{\text{Original weight} - \text{oven-dry weight}}{\text{Oven-dry weight}} \times 100 = \text{moisture content in percent.}$$

4.2.2.2 *Electrical moisture-meter method.*—The moisture content shall be determined in accordance with instructions furnished by the manufacturer of the meter.

5. PREPARATION FOR DELIVERY

5.1 Packing.—Unless otherwise specified, the handles shall be packed as follows.

5.1.1 *"D" handles.*—"D" handles shall be packed in bundles of 12 arranged in 2 layers of 6 each. The bundles shall be formed by placing six handles on edge nesting into one another with the "D" grips in alinement at one end. The second layer shall be formed in the same manner on top of the first layer but reversed end-for-end. Four wood cleats, approximately 8¼ inches long, 2 inches wide, and ¾ inch thick, shall be applied to the bundle; two cleats on top and two on the bottom, each cleat approximately 10 inches from each end of the bundle. Each bundle shall be bound with ¾- x 0.020-inch steel straps conforming to Federal Specification QQ–S–781. One strap shall be applied over each pair of cleats.

5.1.2 *Straight and bent long handles.*—Handles of this description shall be packed in bundles of 24 arranged in 4 layers of 6 each. The grips of the handles in the first two layers shall be at one end and the handles in the third and fourth layers shall be reversed end-for-end. Each bundle shall be bound with two ¾- x 0.020-inch steel straps conforming to Federal Specification QQ–S–781. One strap shall be applied approximately 10 inches from each end.

5.2 Marking.—Unless otherwise specified, each bundle shall bear a label or tag giving the type, class, style, and grade of handles, the number of the contract or order, and the name of the contractor.

6. NOTES

6.1 Ordering data. — Purchasers should specify the type, class, style, and grade of handles required, the type, class, style, and size of tool in which the handles are to be used, and should exercise any desired options offered herein. (See 1.2, 1.3, 5.1, and 5.2.)

6.1.1 The tools referred to in 1.1 may vary in detail depending upon the standards and practice of individual manufacturers. Consequently it may be advisable to include in the

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invitation for bids information concerning the make and model of the tools for which the handles are desired.

6.2 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army.

7.2.1 *Marking.*—Shipping containers for the Army shall be marked in accordance with U. S. Army Specification 100–2, Standard Specification for Marking Shipments by Contractors.

(Copies of specifications required for Army purchases may be obtained upon application to the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.)

7.3 Navy.

7.3.1 *Applicable specifications and other publications.*

Navy Department Specification:

General Specifications for Inspection of Material.

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.

(Copies of Federal specifications required for Navy purchases, Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Commanding Officer, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol or number should be stipulated when requesting copies of specifications.)

7.3.2 *Marking.*—Unless otherwise specified in the contract or order, bundles shall be marked in accordance with the requirements of the Navy Shipment Marking Handbook.

NAVY INTEREST: *MC*ShSY

7.4 Air Force.

7.4.1 *Applicable specification.*

U. S. Army Specification:

94–40645—Marking; Exterior, Domestic and Export Shipment, by Contractors.

(Copies of specifications required for Air Force purchases may be obtained from the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton 1, Ohio.)

7.4.2 *Interior packages.*—Interior packages shall be durably and legibly marked with the following information in such a manner that

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the marking will not become damaged when packages are opened:

Nomenclature
Fed. Spec. NN–H–81a
Manufacturer's part number
Order number
Stock number
Date of manufacture
Quantity
Name of contractor (if not the same as the manufacturer).

7.4.3 *Shipping containers.*—Unless otherwise specified in the contract or order, shipping containers for the Air Force shall be marked in accordance with the requirements of U. S. Army Specification 94–40645.

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

For sale by the Superintendent of Documents, Washington 25, D. C. Price 5 cents.

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	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
3b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	7b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
7c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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